 for development of a model that predicts lymph node involvement. 1

RESULTS

Patient and cervical cancer characteristics are included in Table 1 with 54% of patients with lymph node metastases. Based on primary tumor pathology, 86% of patients had squamous cell carcinoma, and the remainder had either adenocarcinoma or adenosquamous. Using the training set of 57 samples, genes that were differentially expressed between LN+ patients and LN- were identified. 22 genes showed 1.5 or greater fold difference in expression between the groups with FDR 0.01 (Table 2). A heatmap of the differentially expressed genes is shown in Figure 1. Out of the 22 differentially expressed genes, 5 were identified as potential candidates for the predictive model using forward selection: BIRC2, BIRC3, CD300LG, FAM196A, and PHYHIPL. 2 of these genes, FAM196A and PHYHIPL, have very limited gene annotation or relevant clinical background and were therefore excluded from model development. The final three genes, BIRC2, BIRC3, and CD300LG, were included in the training set Random Forest model. The subset of BIRC3 and CD300LG showed the best test set accuracy, 88.2%, and was therefore selected as the final model. In the training set, BIRC3 (Baculoviral IAP Repeat Containing 3) was upregulated 1.7-fold in LN+ patients. CD300LG (Nepmucin), on the other hand, was downregulated 1.5- fold in LN+ patients. Table 3 compares the accuracy of the various models tested, while Figure 2 shows the ROC curve comparison of the 4 models. Figure 3 shows the Random Forest decision surface and a decision tree using the 2 genes with the training set. Alternative models were developed using the final 2 genes with different algorithms, but these models showed slightly lower performance than the Random Forest model. A Support Vector Machine with Radial Basis Function (RBF) kernel had 94.4% ROC-AUC, and a Gaussian Naïve Bayes model had 84.7% ROC-AUC. The Random Forest model correctly classified 88.2% of the test set patients, n = 17.1 All 8 of the LN+ patients were correctly classified; out of the 9 LN- patients, 7 were correctly classified. The test set ROC- AUC was 98.6%, 95% CI [66.7%, 100%]. Sensitivity for detection of LN involvement was 100%, and specificity was 77.8%. Precision was 80.0% and Negative Predictive Potential was 100%.

DISCUSSION

Lymph node involvement represents one of the most significant determinants of cervical cancer prognosis and impacts treatment approach [3–6]. Our pilot study shows that lymph node involvement in cervical cancer can be predicted with RNA-seq data, and our two-gene lymph node predictive signature showed 88% predictive accuracy when evaluated in a separate cohort. Cervical cancer lymph node involvement can be determined by surgical staging or imaging, such as FDG- PET/CT or MRI. Several studies suggest that lymph node status on FDG-PET is a more significant predictor of disease outcome than clinical FIGO stage [6–8]. The presence of lymph nodes also influences treatment decisions, such as the need for adjuvant chemoradiation after surgery or the extent of the radiation field and dose for definitive chemoradiation treatment. Currently, no simple lab test that accurately predicts cervical cancer lymph node involvement exists. Additionally, the incidence of cervical cancer is increasing in developing countries with limited health care resources for surgery, imaging, and treatment. A simple pathologic tool that could help stratify patients based on predicted lymph node status could be particularly beneficial for determining the best utilization of treatment resources. If treatment or radiation resources are particularly limited, knowing which patients have no lymph nodes could help identify the patients to treat with a definitive or curative approach versus those to treat more palliatively. Alternatively, if imaging resources are limited but surgical and/or radiation resources are available, it might be that patients with a biomarker predicting for lymph nodes undergo surgical lymph node resection and/or extended field radiation or additional chemotherapy. A limited number of microarray gene expression studies have been performed involving cervical cancer, and a few groups have attempted to identify a gene expression signature that can predict lymph node involvement in cervical cancer [9–12]. Many of these studies were relatively small and lacked a validation cohort. Grigsby et al., for example, included 8 cervical cancer patients, 3 with supraclavicular metastases, and identified 75 out of 12,000 genes with at least 3-fold differential expression to create a 12 gene signature to distinguish the groups [9]. Biewenga et al. evaluated tumor samples from 35 patients (16 with lymph node metastases) and found that 5 genes with differential expression had a prediction accuracy of 64.5% [10]. Using 43 primary cervical cancer samples (16 with lymph node metastases), Kim et al. created a lymph node prediction model using 156 genes with a prediction accuracy of 77% [11]. While Huang and colleagues did include a validation cohort, they evaluated early stage cervical cancer patients undergoing hysterectomy with fairly low rates of lymph node metastases [12]. In contrast, our study includes a range of stages, a greater proportion of advanced stage, a higher proportion with lymph node metastases, and also used deep sequencing with RNA-seq that assayed over 16,000 genes, as opposed to microarrays with fewer than 1,500 genes. Unfortunately, these different microarray studies for cervical cancer had minimal overlap in significant genes, suggesting that a new, more